

skAlnet Edge-Compute v1.5

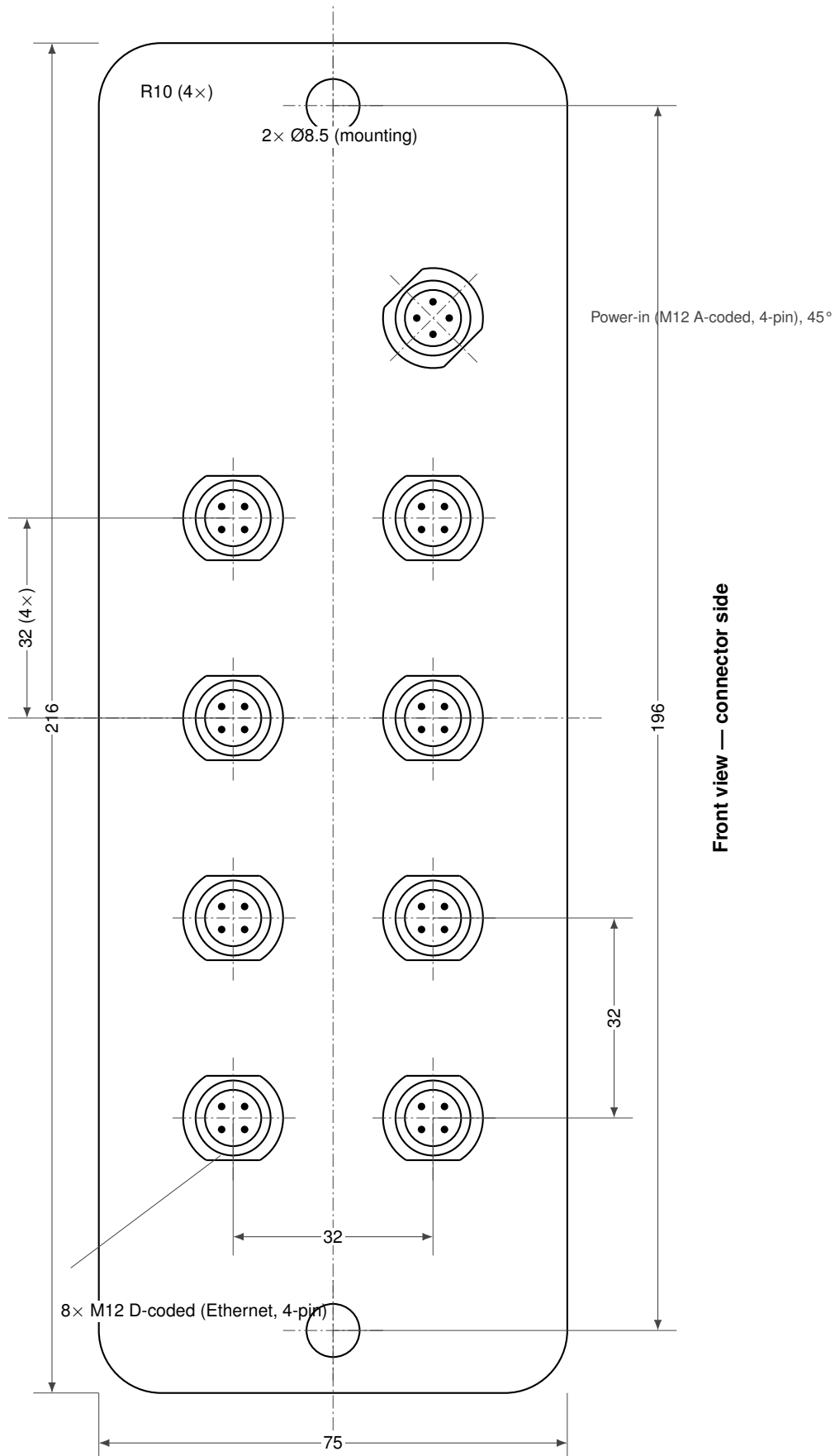
Programmable M12-PoE switch, router, and Linux compute module for industrial edge data acquisition



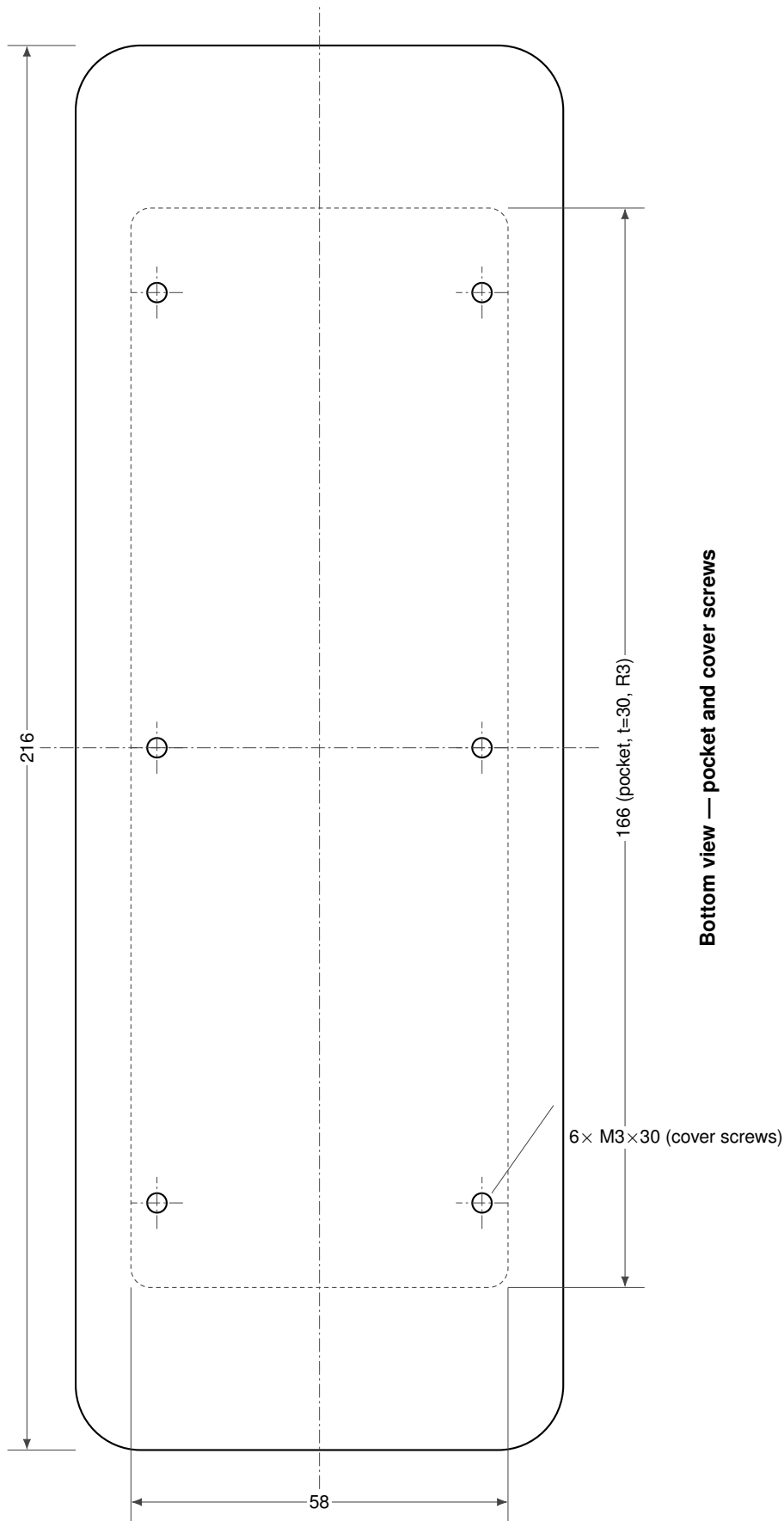
Description. The skAlnet Edge-Compute is Auto-Intern GmbH’s universal data-acquisition and monitoring platform. An anodized aluminum enclosure, machined from a single block, combines an 8-port switch with seven external M12 PoE+ LAN ports (D-coded), a separate M12 WAN port (D-coded), and a swappable compute module (Raspberry Pi CM pin-compatible). All connections use sealing M12 connectors; the device operates under water at up to 1 bar. Measurement data is buffered, pre-processed, and analyzed on the device; only data explicitly configured for forwarding reaches the upstream network.

Compute		Power	
Module	Raspberry Pi CM4 (standard); CM5 or NVIDIA Jetson on request, pin-compatible, swappable	Input	48 V to 72 V DC via M12 A-coded (4-pin, TE T4140012041-000)
CPU	8 cores, 64-bit ARM, 1.5 GHz	PSU	recommended $\geq 100\text{ W}$
RAM	8 GB LPDDR4-3200	Minimum	25 W (compute only, no PoE loads)
Storage	32 GB eMMC	Environmental & mechanical	
Operating system	Yocto Linux with documented SBOM (EU Cyber Resilience Act); encrypted OTA updates or offline via USB image	Enclosure	anodized aluminum, machined from solid, shockproof
Networking		Ingress protection	IP67, all ports and back plate sealed; operation under water up to 1 bar
WAN	1 $\times$ M12 D-coded (4-pin), Giga-bit Ethernet, DHCP client	Temperature	-25°C to 70°C
LAN	7 $\times$ M12 D-coded (4-pin, Ethernet) with PoE+, internally connected to an 8-port switch (DHCP server, 192.168.199.1/24)	Dimensions	216 $\times$ 75 $\times$ 32 mm (W $\times$ D $\times$ H); overall depth incl. connectors approx. 51 mm
PoE	IEEE 802.3at (PoE+) on all 7 LAN ports; typical devices draw up to approx. 10 W	Front plate	8 $\times$ M12 D-coded (7 $\times$ PoE LAN, 1 $\times$ WAN) in a 4 $\times$ 2 grid, 1 $\times$ M12 A-coded power-in; custom branding available
Cabling	1 certified Cat-5e M12 cable per device	Software	
Data interface	Live data via WebSocket (port 8001, JSON, ns timestamps)	Access	SSH (port 22022); custom applications in C++ or Python, Docker deployment; minimal examples on GitHub
Upstream APIs	REST, WebSocket, webhooks, MQTT, OPC UA, Modbus-TCP, EPICS, gRPC, AMQP, CoAP, SNMP, Prometheus, InfluxDB, SFTP/rsync	Remote support	optional manufacturer remote maintenance via encrypted access, can be disabled
Capacity	250+ concurrent channels proven in production	Updates	encrypted OTA updates via the WAN port or offline via USB image
		Field references	
		DB Netz DIANA (switch-point motor monitoring) · Kelvion (IR fouling detection) · Kurtz Ersä / Global Point HORUS (soldering process monitoring) · RUB/GSI-FAIR PANDA (luminosity detector) · MSU CBE (river impedance spectroscopy) · NexuFed AI / RUB (federated pump monitoring)	

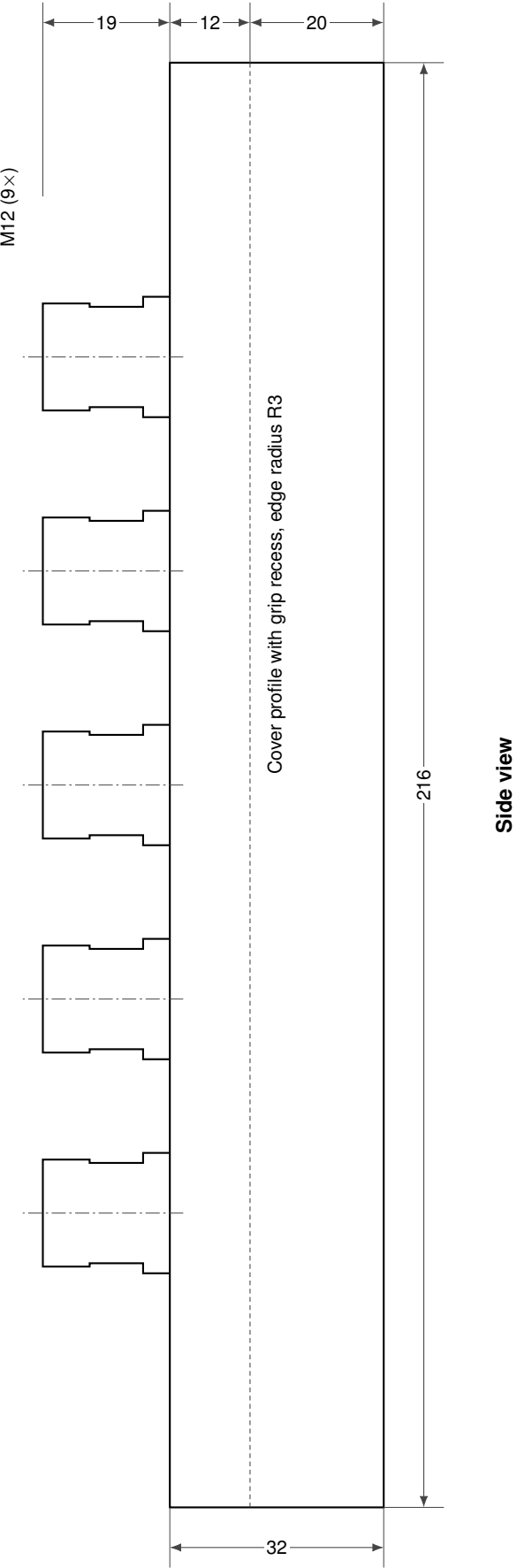
Manufacturer: Auto-Intern GmbH · Herner Str. 299, Building B29, 44809 Bochum, Germany · info@auto-intern.de · +49 234 9345 1121
Price: from 260 € / \$299 net · **Documentation:** edge-compute.skainet.io

Dimensioned Drawing — Enclosure Edge-Compute v1.5All dimensions in mm · General tolerances ISO 2768-mK · Material: Al, anodized · **Scale 1:1** · Sheet 1/4: front view

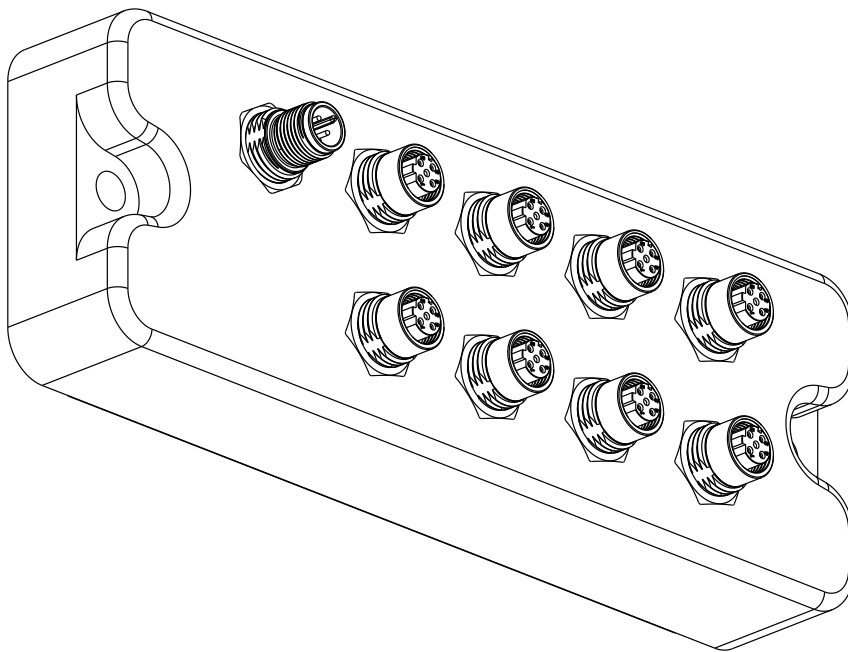
All dimensions in mm · **Scale 1:1** · Sheet 2/4: bottom view



All dimensions in mm · **Scale 1:1** · Sheet 3/4: side view



Sheet 4/4: isometric view of the compute module, connector side (not to scale)



All connectors M12, IP67: 8 × D-coded (7 × PoE LAN, 1 × WAN, 4-pin Ethernet), pitch 32 × 32; 1 × A-coded power-in (4-pin), rotated 45°. Cover fastening: 6 × M3 × 30. Subject to change without notice.